

by Julian Chang

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$$h(K) = \frac{g(K)}{\text{IRR}(K)}$$

for $K_1 < S_T < K_2$,

$$g(K) = K - K_1 \quad g'(K) = 1 \quad g''(K) = 0$$

$$h'(K) = \frac{\text{IRR}(K) - g(K)\text{IRR}'(K)}{\text{IRR}(K)^2} \quad h''(K) = \frac{-\text{IRR}''(K)g(K) - 2\text{IRR}'(K)}{\text{IRR}(K)^2} + \frac{2\text{IRR}'(K)^2g(K)}{\text{IRR}(K)^3}$$

for $S_T > K_2$,

$$g(K) = K_2 - K_1 \quad g'(K) = g''(K) = 0$$

$$h'(K) = \frac{-g(K)\text{IRR}'(K)}{\text{IRR}(K)^2} \quad h''(K) = \frac{-\text{IRR}''(K)g(K)}{\text{IRR}(K)^2} + \frac{2 \cdot \text{IRR}'(K)^2g(K)}{\text{IRR}(K)^3}$$

Integrate from K_1 to since less than K_1 there is no payoff

$$\begin{aligned} V_0 &= \int_0^\infty h(K) \frac{\partial^2 V^{\text{pay}}(K)}{\partial K^2} dK \\ &= \int_{K_1}^{K_2} h(K) \frac{\partial^2 V^{\text{pay}}(K)}{\partial K^2} dK + \int_{K_2}^\infty h(K) \frac{\partial^2 V^{\text{pay}}(K)}{\partial K^2} dK \\ &= \left[h(K) \frac{\partial V^{\text{pay}}(K)}{\partial K} \right]_{K_1}^{K_2} - \int_{K_1}^{K_2} h'(K) \frac{\partial V^{\text{pay}}(K)}{\partial K} dK \\ &\quad + \left[h(K) \frac{\partial V^{\text{pay}}(K)}{\partial K} \right]_{K_2}^\infty - \int_{K_2}^\infty h(K) \frac{\partial V^{\text{pay}}(K)}{\partial K} dK \\ &= \cancel{h(K_2) \frac{\partial V^{\text{pay}}(K_2)}{\partial K}} - \cancel{h(K_1) \frac{\partial V^{\text{pay}}(K_1)}{\partial K}} - \int_{K_1}^{K_2} h'(K) \frac{\partial V^{\text{pay}}(K)}{\partial K} dK \\ &\quad + \cancel{h(\infty) \frac{\partial V^{\text{pay}}(\infty)}{\partial K}} - \cancel{h(K_2) \frac{\partial V^{\text{pay}}(K_2)}{\partial K}} - \int_{K_2}^\infty h(K) \frac{\partial V^{\text{pay}}(K)}{\partial K} dK \\ &= - \left[\cancel{h'(K_2) V^{\text{pay}}(K_2)} - h'(K_1) V^{\text{pay}}(K_1) - \int_{K_1}^{K_2} h''(K) V^{\text{pay}}(K) dK \right] \\ &\quad - \left[\cancel{h'(\infty) V^{\text{pay}}(\infty)} - \cancel{h'(K_2) V^{\text{pay}}(K_2)} - \int_{K_2}^\infty h''(K) V^{\text{pay}}(K) dK \right] \\ &= h'(K_1) V^{\text{pay}}(K_1) + \int_{K_1}^{K_2} h''(K) V^{\text{pay}}(K) dK + \int_{K_2}^\infty h''(K) V^{\text{pay}}(K) dK \\ V_0 &= \frac{1}{\text{IRR}(K_1)} V^{\text{pay}}(K_1) + \int_{K_1}^{K_2} h''(K) V^{\text{pay}}(K) dK + \int_{K_2}^\infty h''(K) V^{\text{pay}}(K) dK \end{aligned}$$

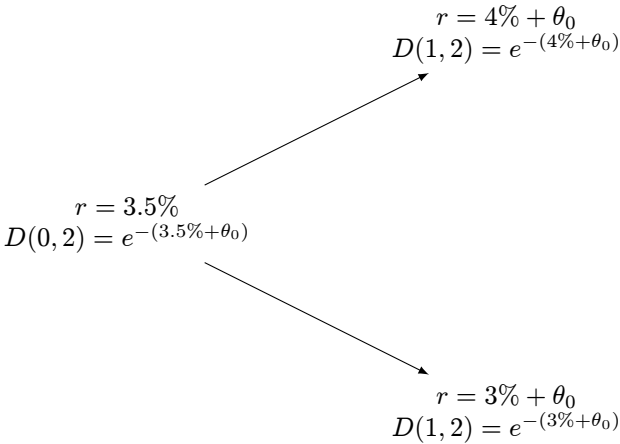
wehre $g(K_1) = 0, h'(K_1) = \frac{1}{IRR(K_1)}$

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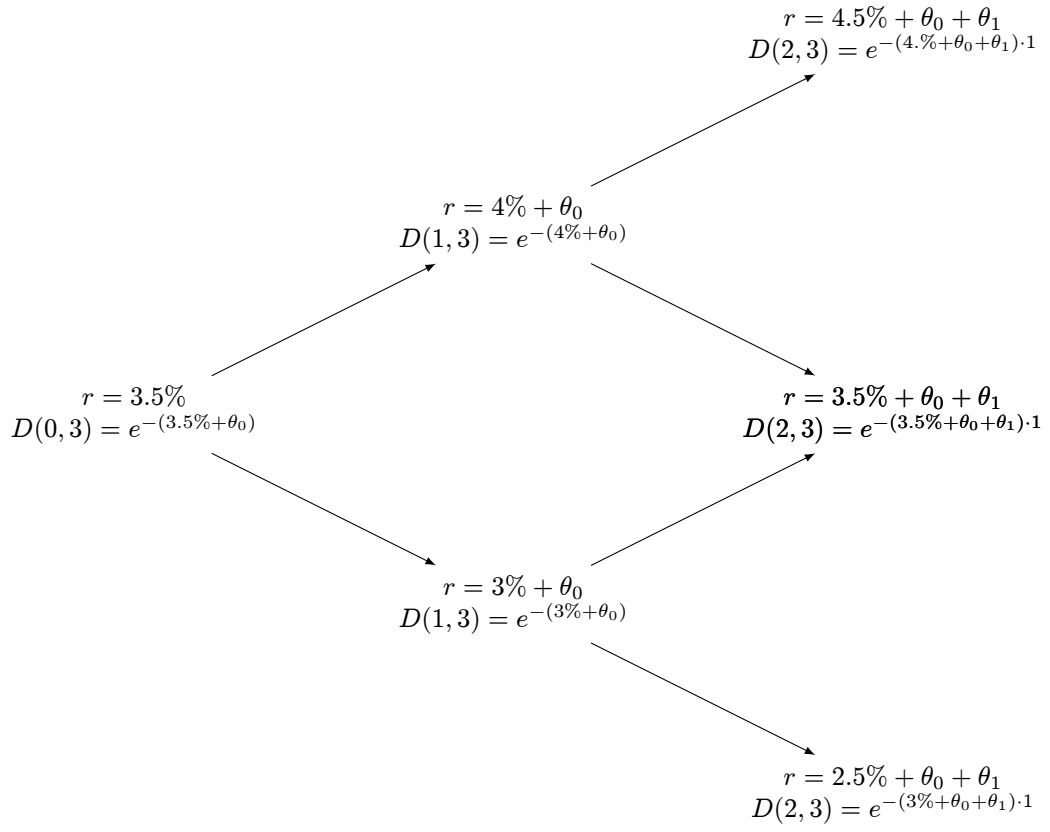
$$\begin{aligned} r_t &= r_0 + \int_0^t \theta(s)ds + \int_0^t \sigma dW_s^* \\ \int_0^t r_u du &= \int_0^t r_0 du + \int_0^t \int_0^u \theta(s)dsdu + \int_0^t \int_0^u \sigma dW_s^* du \\ &= r_0 t + \int_0^t \theta(s)(t-s)ds + \int_0^t \sigma(t-s)dW_s^* \\ \mathbb{E}\left[\int_0^t r_u du\right] &= r_0 t + \int_0^t \theta(s)(t-s)ds \\ \mathbb{V}\left[\int_0^t r_u du\right] &= \int_0^t \sigma^2(t-s)^2 dt = \frac{1}{3}\sigma^2 t^3 \end{aligned}$$

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R(0,1) = 3.5%



$$\begin{aligned} D(0, 2y) &= D(0, 1y) \times E^*[D(1y, 2y)] \\ &= D(0, 1y) \times \left[\frac{1}{2} \times e^{-(\theta_0 + 0.04)} + \frac{1}{2} \times e^{-(\theta_0 + 0.03)} \right] \\ \Rightarrow \theta_0 &= 0.010783 \quad \triangleleft \end{aligned}$$



$$D_u(1, 3) = e^{-(\theta_0 + 0.04)} \times \left[\frac{1}{2} \times e^{-(\theta_0 + \theta_1 + 0.045)} + \frac{1}{2} \times e^{-(\theta_0 + \theta_1 + 0.035)} \right]$$

$$D_d(1, 3) = e^{-(\theta_0 + 0.03)} \times \left[\frac{1}{2} \times e^{-(\theta_0 + \theta_1 + 0.035)} + \frac{1}{2} \times e^{-(\theta_0 + \theta_1 + 0.025)} \right]$$

$$D(0, 3y) = D(0, 1y)E^*[D(1, 3)] = D(0, 1y) \times \left[\frac{1}{2} \times D_u(1, 3) + \frac{1}{2} \times D_d(1, 3) \right]$$

$$\Rightarrow \theta_1 = -0.010301 \quad \triangleleft$$